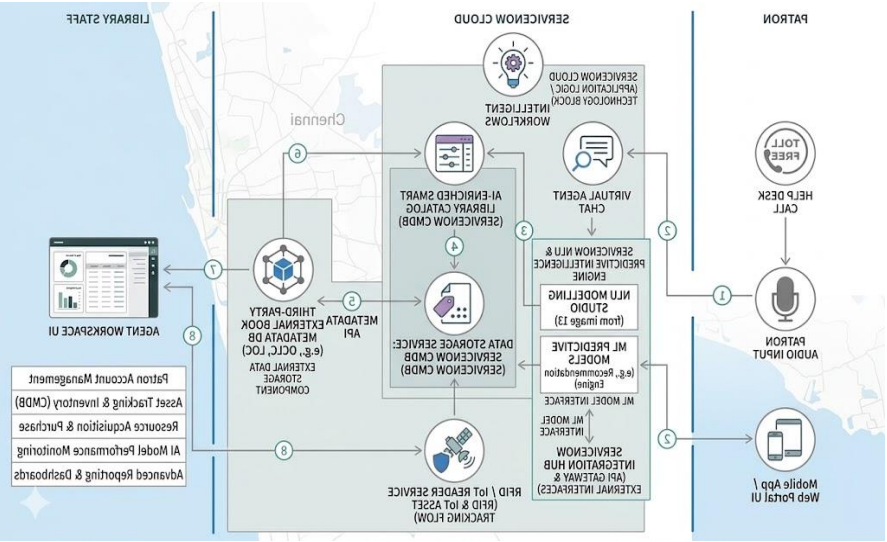
Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	05 Apr. 26
Team ID	SWTID-2026-6042
Project Name	Smart Library Request Workflow in ServiceNow
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2



Smart Library: Architecture Highlights

- **Logic (Cloud): ServiceNow Flow Designer** & **Virtual Agent** (Core Application Logic).
- **Infrastructure: ServiceNow Cloud Instance** (Back-end) vs. **Local IoT/Mobile** (Edge inputs).
- **External APIs: Integration Hub** connecting to third-party **Book Metadata DBs** (REST/SOAP).
- **Data Storage: ServiceNow CMDB** (Asset/Inventory) & **Foundation Tables** (Patron Data).
- **ML Interface: Predictive Intelligence** & **NLU Studio** for recommendation/voice models.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Interaction points for patrons and librarians.	ServiceNow Service Portal (AngularJS), Mobile Agent App, Virtual Agent.
2.	Application Logic-1	Core borrowing and return workflow automation.	ServiceNow Flow Designer / JavaScript (Business Rules).
3.	Application Logic-2	Natural Language Processing for voice/text search.	ServiceNow NLU (Natural Language Understanding).
4.	Application Logic-3	Automated book recommendations and classification.	ServiceNow Predictive Intelligence (ML).
5.	Database	Relational data for patrons and lending rules.	ServiceNow Internal MariaDB (Relational).
6.	Cloud Database	Asset and metadata repository.	ServiceNow CMDB (Configuration Management Database).
7.	File Storage	Digital resource storage (e-books, manual PDFs).	ServiceNow Attachment Table / Managed Cloud Storage.
8.	External API-1	Retrieval of global book metadata/titles.	Google Books API / OCLC WorldCat API (REST).
9.	External API-2	User authentication and identity verification.	Azure AD / Okta (SAML 2.0 / OAuth 2.0).
10.	Machine Learning Model	Pattern recognition for inventory and fraud detection.	ServiceNow AI Search & Clustering Models.
11.	Infrastructure (Server / Cloud)	Multi-instance cloud architecture deployment.	ServiceNow Cloud (AOB), Mid Server (for Local IoT/RFID).

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Web components and client-side scripting for the Portal.	AngularJS, Bootstrap, Node.js (MID Server).
2.	Security Implementations	Role-based access (RBAC), multi-factor authentication, and data encryption.	SAML 2.0, OAuth 2.0, AES-256 Encryption, ACLs (Access Control Lists).

S.No	Characteristics	Description	Technology
3.	Scalable Architecture	Multi-instance architecture that separates web, application, and data layers.	Advanced High Availability (AHA), ServiceNow Multi-Instance Cloud.
4.	Availability	Geography-redundant data centers with automated failover.	Load Balancers, N+1 Redundancy, ServiceNow AHA Architecture.
5.	Performance	Optimized query execution and edge-side processing for low latency.	Semiconductor Caching, Edge Encryption, ServiceNow Integration Hub.